

Cellular Biophysics (BIO516) – Mid Sem

Marks: 20, Time: 1 hr

Date: 26/02/2026

Instructions to candidates

1. This paper consists of four sections: Section A (Multiple Choice: Q1-Q7), Section B (Fill in the Blanks: Q8-Q11), Section C (Numerical Problems: Q12 & Q13), and Section D (Short Subjective Questions: Q14-Q16)

2. Answer all the questions

3. Use of calculator is permitted.

4. Standard Constants:

- Gas Constant (R) = $8.315 \text{ J/(mol} \cdot \text{K)}$
- Faraday's Constant (F) = $96,485 \text{ C/mol}$
- Temperature (T) = 298 K (unless specified)

Section A

[Marks: 7 X 1]

Q1 The Debye length (d_D) represents the distance at which the potential in the diffuse layer drops to $1/e$ of its value at the OHP. How does d_D change as the ionic strength of the medium increases?

- A) d_D increases linearly.
- B) d_D remains unchanged.
- C) d_D decreases.**
- D) d_D increases as the square of the concentration.

Q2 According to Fick's First Law, if the thickness of the membrane doubles while the difference in solute concentration in two compartments across the membrane remains constant, the Net Flux (J) will:

- A) Double.
- B) Remain unchanged.
- C) Decrease by half.**
- D) Increase by a factor of 4.

Q3 In the context of the Behn equation for diffusion potential, what happens when the mobilities of the cation (U_C) and anion (U_A) are equal?

- A) The diffusion potential reaches its maximum positive value.
- B) The diffusion potential is zero.**
- C) The membrane potential becomes the average of the two Nernst potentials.
- D) The system enters a state of active transport.

Q4 In the Eisenman selectivity sequences, which sequence is characteristic of a high field strength site (where the site-ion interaction dominates over hydration)?

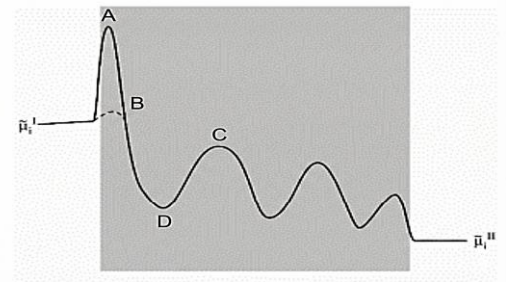
- A) $\text{Cs}^+ > \text{Rb}^+ > \text{K}^+ > \text{Na}^+ > \text{Li}^+$
- B) $\text{Li}^+ > \text{Na}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+$**
- C) $\text{K}^+ > \text{Na}^+ > \text{Li}^+ > \text{Rb}^+ > \text{Cs}^+$
- D) $\text{Na}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+ > \text{Li}^+$

Q5 The Lipid Bilayer functions as a _____ while the Ion Channels function as _____.

- A) Resistor; Capacitors
- B) Capacitor; Resistors**
- C) Battery; Conductors

D) Inductor; Batteries

Q6. Identify A, B, C & D in the RHS figure, based on the energy profile for an ion passing through a channel.



A) A: Closed on gate barrier, B: Open on gate barrier

B) C: Well at binding site, D: Intense Barrier

C) A: Open on gate barrier, B: Closed on gate barrier

D) C: Intense Barrier, D: Well at binding site

Q7 For an ion channel gating process where the closing rate α is given by $\alpha = Be^{A\Delta\psi}$, an increase in the membrane potential $\Delta\psi$ will:

A) **Increase the mean duration of the open state.**

B) **Decrease the mean duration of the open state.**

C) Have no effect on the transition kinetics.

D) Cause the channel to degrade into the lipid matrix.

(*** (A) is correct if you use the convention of negative membrane potential (inside negative) and consider hyperpolarization as an increase in membrane potential (i.e. from -90 mV to -100 mV, closing rate decreases).

(B) is correct if you use the convention of negative membrane potential (inside negative) and consider an increase in membrane potential from -90 mV implies going toward more positive direction (e.g., -70 mV, i.e., depolarization). Here closing rate increases with increase in membrane potential).

Section B

[Marks: 4 X 1]

Q8. when the temperature of the environment increases, the viscosity would _____, but fluidity would _____. (Answer: **decrease/increase**)

Q9. The amino acids _____ and _____ are positively charged at physiological pH (i.e., pH 7). (Ans: any two of **Arginine, Histidine, lysine**)

Q10 According to the GHK constant-field equation, if the membrane is permeable to only one ion, the resting potential reduces to the _____ potential for that ion. (Ans: **Nernst**)

Q11. In the Teorell flux equation, flux is defined as the product of Mobility, Concentration, and the _____. (Ans: **Driving Force**)

Section C

[Marks: 2 X 2]

Q12. Calculate the resting membrane potential ($\Delta\psi_r$) at 25°C using the GHK equation. **given:** $[K^+]_e = 5$ mM, $[K^+]_i = 150$ mM, $[Na^+]_e = 140$ mM, $[Na^+]_i = 15$ mM. Assume $P_{Na}/P_K = 0.02$ and the membrane is impermeable to Cl^- .

Ans: $2.303RT/F$ at 25 °C = 59 mV

So, the resting membrane potential

$$\Delta\Psi_r = 59 \log_{10} \frac{[K^+]_e + (\frac{P_{Na}}{P_K})[Na^+]_e}{[K^+]_i + (\frac{P_{Na}}{P_K})[Na^+]_i}$$

$$\Delta\psi = 59 \log_{10} \frac{5+0.02(140)}{150+0.02(15)} = 59 \log_{10} \frac{7.8}{150.3} \approx -75.8 \text{ mV}.$$

Q13. Calculate the Free Energy change for transporting (ΔG_t) 1 mole of Na^+ out of the cell against its electrochemical gradient at 25° C. Given: $Na_{in}^+ = 14 \text{ mM}$, $Na_{out}^+ = 140 \text{ mM}$, membrane potential (V_m) = -70 mV (Inside is negative relative to outside), Faraday Constant $F = 96,500 \text{ J/(V.mol)}$.

Ans. Chemical Term: Against chemical gradient

$$\Delta G_t \text{ (due to chemical gradient)} = RT \ln \left(\frac{[C_{final}]}{[C_{initial}]} \right) = RT \ln(10) = +5.7 \text{ kJ/mol. (Uphill concentration).}$$

Electrical Term: against electrical gradient (i.e., work done in transporting + charge from the side of the membrane having -ve electrical potential to the side of the membrane having relatively +ve electrical potential). (**convention:** transmembrane potential -70 mV (negative inside))

$$\Delta G_t \text{ (due to electrical gradient)} = -ZF(-70 \text{ mV}) = -(+1)(96,500 \text{ J/V.mol})(-0.07 \text{ V}) = +6,755 \text{ J} = \text{approx } +6.8 \text{ kJ/mol.}$$

$$\text{Total: } \Delta G_{t, \text{total}} = +5.7 + 6.8 = +12.5 \text{ kJ/mol.}$$

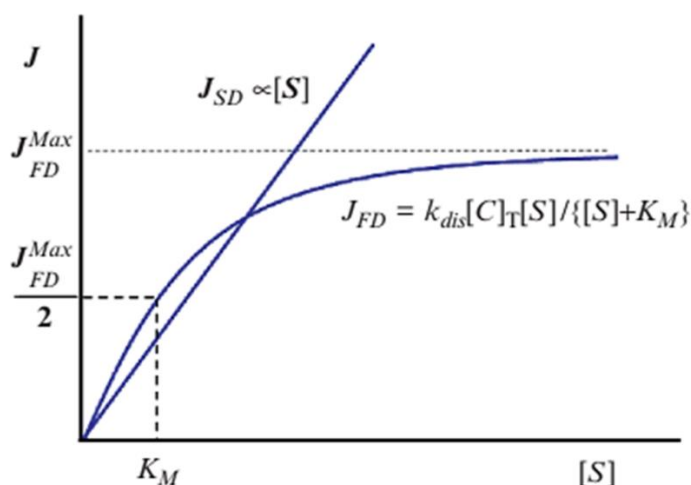
Section D

Q14 Explain the “Hydrophobic Effect” and its role in the spontaneous formation of lipid bilayers. Why is the increase in water entropy ($\Delta S > 0$) critical to this process? [Marks: 1]

Ans: Hydrophobic effect: clustering of hydrophobic region/tail of molecules to minimize their interface with water molecules in an aqueous medium. This is driven by entropy; clustering non-polar region/tails releases ordered water “cages,” increasing total entropy ($\Delta S > 0$)

Q15 Draw the graph of simple diffusion and carrier-mediated diffusion of solutes across a membrane. Compare their flux, temperature sensitivity, and saturability. [Marks: 2]

Ans: The flux dependence on the concentration of solute in carrier-facilitated diffusion (FD) and simple diffusion (SD).



	Simple diffusion	Facilitated diffusion
Flux	Linear with the concentration of the solute.	Attains an asymptotic limit (i.e. saturation) at higher concentrations of solute.
Temperature sensitivity	Flux is proportional to temperature (As the diffusion coefficient is proportional to the temperature). The temperature coefficient Q ₁₀ is 2-3.	Flux depends on temperature in an exponential manner ($A \exp(-E_a/T)$). Q ₁₀ is more than 3.
Saturability	Doesn't attain saturation with solute concentration.	Attains an asymptotic limit at a higher concentration of solute due to the saturation of available carrier.

Q16 Differentiate between Electrolyte and non-electrolyte diffusion in terms of the form of driving force, dependence on transmembrane potential and type of transporters. [Marks: 2]

Ans:

	Non-electrolyte diffusion	Electrolyte diffusion
Driving force	Concentration gradient (i.e., chemical gradient).	Concentration gradient + membrane potential gradient (i.e., electrochemical gradient).
dependence on transmembrane potential	None.	yes.
type of transporters	Carriers.	Carriers and channels.